



Machine Learning Roadmap

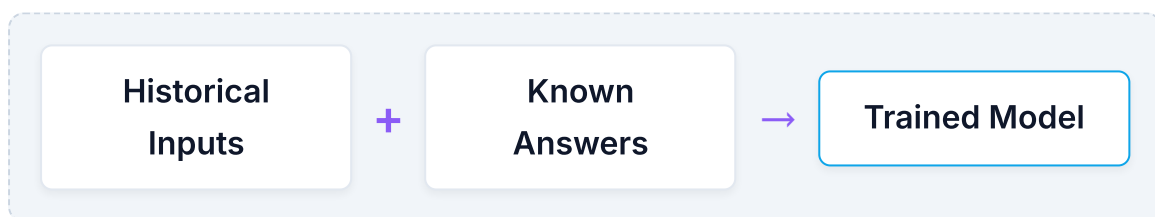
PART 2: TYPES OF MACHINE LEARNING 🐍 📊

Machine Learning isn't a single monolithic technique. Different problems require entirely different learning paradigms. Understanding these core types is the most critical step before you write a single line of code or train a model.



1. Supervised Learning

In supervised learning, the model learns from **labeled historical data**. The algorithm is provided with both the input data and the exact correct answer (the target) it should try to predict.



Example:

- 2 hours of study → *Score: 50*
- 6 hours of study → *Score: 80*

Regression

Used when the target output is a **continuous numerical value**.

Use Cases:

- 🏠 Predicting house prices (\$250,500.50)
- 🌡️ Forecasting daily temperature
- 📦 Predicting exact product demand

Classification

Used when the target output is a **discrete category or class**.

Use Cases:

- 📧 Spam vs. Not Spam
- 💳 Fraudulent vs. Legitimate
- 🐱 Image recognition (Cat vs. Dog)

⚠️ **Crucial Beginner Distinction:** Do not assume that just because the output is a number, it is Regression. For example, predicting a *Zip Code* or a *Customer Rating* (1, 2, 3, 4, 5) involves numbers, but they represent distinct categories. These are **Classification** problems!

2. Unsupervised Learning

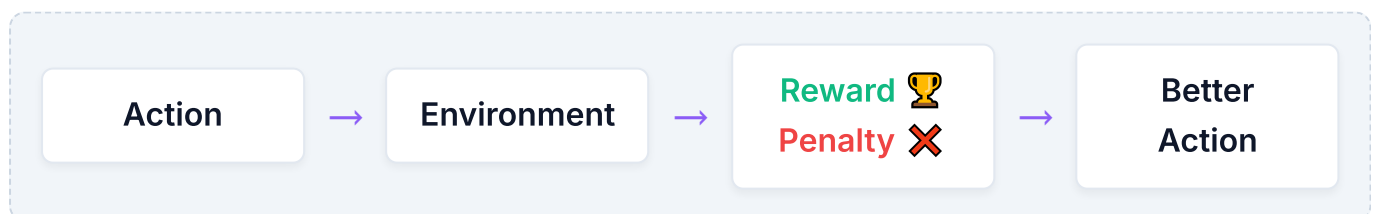
In unsupervised learning, the data does **not** contain predefined targets or labels. The model acts as a detective, attempting to discover hidden structures, patterns, or groupings within raw data autonomously.

Core Unsupervised Techniques:

Technique	Description & Example
 Clustering	Groups similar observations together based on traits. <i>Example: Grouping customers into "Frequent Buyers," "Bargain Hunters," and "Dormant Users" based on raw purchase data.</i>
 Association Rules	Discovers rules that describe large portions of your data. <i>Example: Market Basket Analysis ("Customers who bought a laptop also bought a wireless mouse 80% of the time").</i>
 Dimensionality Reduction	Compresses data while retaining essential information to remove noise or prepare for visualization.

3. Reinforcement Learning

Reinforcement Learning moves away from static datasets. Instead, an **Agent** interacts with a dynamic **Environment** and learns purely through trial, error, and feedback.



Common Applications: Training chess engines (AlphaZero), autonomous driving systems, and complex robotic navigation.

4. Decision Matrix: Which Type Should You Use?

Before writing code, ask yourself these diagnostic questions:

Diagnostic Question	Recommended Approach
? Do I have a historical dataset with a known target to predict?	➡ Supervised Learning
↳ <i>Is that target a continuous number? (e.g., Price, Salary)</i>	➡ Regression
↳ <i>Is that target a specific category? (e.g., Yes/No, Red/Blue)</i>	➡ Classification
? Am I trying to find hidden groups or segments in raw, unlabeled data?	➡ Clustering (Unsupervised)
? Does an AI agent need to learn optimal behavior through trial and error?	➡ Reinforcement Learning

5. Knowledge Check Exercise

Identify the most appropriate ML approach for each real-world scenario:

- 1 Predict the resale value of a used car based on mileage.
- 2 Determine whether a live credit card transaction is fraudulent.
- 3 Segment an email list based on historical reading behavior.
- 4 Forecast inventory requirements for next month.
- 5 Train an algorithm to balance a pole on a moving cart.

Answers: 1. Regression 2. Classification 3. Clustering
4. Regression (Time-Series) 5. Reinforcement Learning

Professional Tip

The most common beginner mistake is asking, "Which algorithm should I use?" (e.g., Random Forest vs. SVM). Always start by asking, "What kind of problem am I solving?" Once you classify your problem mathematically (Regression, Classification, or Clustering), the pool of applicable algorithms instantly narrows down for you.